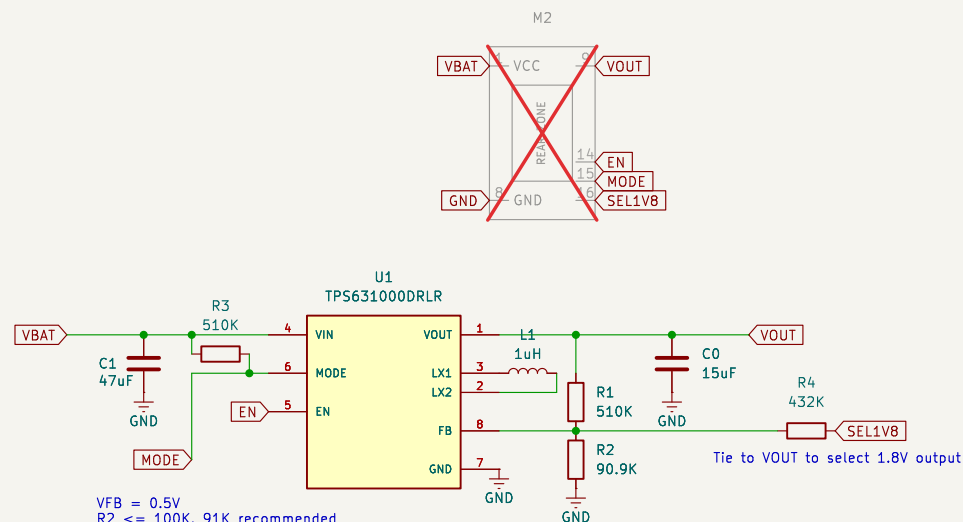


Mega65Modular Low current DC/DC v0.1
gardners
License: CERN-OHL-S v2
Year: 2025

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VFB = 0.5V
R2 <= 100K. 91K recommended

$R1 = R2 \times (VOUT / VFB - 1)$
1.8V: $R1 = R2 \times (1.8 / 0.5 - 1) = R2 \times 2.6 = 234K$
3.3V: $R1 = R2 \times (3.3 / 0.5 - 1) = R2 \times 5.6 = 511K$

So SEL1V8 should tie FB to VOUT with a resistor that in parallel to 511K yields 234K.

$1/RP = 1/R + 1/R$
 $1/260K = 1/R + 1/511K$
 $1/R = 1/234K - 1/511K$
 $1/R = \sim 1/436K$
 $R = \sim 436K$

Actual values = 90.9K, 510K, so VFB = 0.5V at junction means
 $V_{OUT} = (1 + 510K/90.9K) * 0.5V = 3.305V$
 And with SEL1V8 tied to VOUT:
 $1/R = 1/510K + 1/432K = 233.8K$
 $V_{OUT} = (1 + 233.8K/90.9K) * 0.5V = 1.79V$
 Close enough...

3D top side